

The Song and the Strange Attractor

Can the brain spin out a stereotyped sequence — a birdsong, a memory, a readiness to move — out of pure deterministic chaos? In this duel the idea of "chaotic itinerancy" met a hard test: everything it explains turns out to have a simpler explanation, and everything it predicts collides with the measurements.

Date

8 June 2026

Topic

Whether real neural sequences (zebra-finch song, hippocampal replay, cortical preparatory sequences) are generated by deterministic chaotic itinerancy with a positive Lyapunov exponent, rather than by stable, noise-driven sequential dynamics

Claim type

Numeric — backed only by the attached reproducible compute bundle

Sides

Aramis — defending the thesis · Atos — refuting it · (public AI-generated sides; engine identities withheld)

Format

twelve alternating moves, the decisive blows simulated or solved in closed form; an independent AI arbiter; an order-swap audit

Status: public adversarial AI stress-test. This is not peer review, not a scientific proof, and not an endorsement of the claim.

Two AI-generated sides argued a frozen claim under a fixed protocol; an independent AI arbiter issued a structured verdict. For numerical claims, the reported numbers come only from the attached reproducible compute bundle. Publication means the duel passed the release gate; it does not promote the claim to discovery status. Passing the release gate means automated structural, integrity and format checks passed — it is not human methodological review, peer review, or approval of the claim's substance.

Epistemic status: scout — exploratory. This issue is a scouting duel, **not an established scientific result**. Its decisive test runs on synthetic, literature-calibrated toy models and a closed-form identity: it probes whether deterministic chaos is *necessary in principle* to explain the songbird numbers, not whether any particular brain recording lives in a given regime. Treat the finding as a lead, not a confirmed claim about real neural data.

Some of the most precise behaviours in biology are **sequences**: a zebra finch sings the same syllables in the same order, thousands of times; the hippocampus "replays" a travelled route; the cortex marches through a fixed series of states before a movement. A long-standing question is what kind of dynamical machine produces them.

Two machines, one behaviour. One candidate is a **stable chain**: each state hands off to the next like falling dominoes, clocked by a slow biophysical variable (neural *adaptation* or synaptic depression) that builds until it crosses a threshold and triggers the next step. The other is **deterministic chaotic itinerancy** — the trajectory wanders along a "strange attractor" with a small positive **Lyapunov exponent** ($\lambda_1 > 0$), the mathematical fingerprint of sensitive dependence on initial conditions: tiny differences blow up over time.

A recent theory paper proved that such chaotic sequences really can exist in a clean class of model networks. The duel took the next, riskier step the paper itself did *not* take: it asked whether *real* brains use that mechanism — and specifically whether the positive Lyapunov exponent is the **source** of the timing variability we measure.

Why this is testable. Birdsong shows a striking split: the **order** of syllables is fixed to better than 99% accuracy, while their **durations** wobble from rendering to rendering with a coefficient of variation (CV) of about 5–15%. "Order rigid, timing loose" is exactly what chaotic itinerancy predicts. The thesis went further: that loose timing *requires* a positive Lyapunov exponent. The falsifier accepted by both sides: the thesis fails if (a) the timing precision actually implies $\lambda_1 \leq 0$, AND (b) a stable "noise + adaptation" model matches the statistics at least as well.

The frozen claim

The thesis Aramis was assigned to defend, fixed before the duel and not edited afterwards (translated):

*"Real neural sequences (zebra-finch song, hippocampal replay, cortical preparatory sequences) are generated by **deterministic chaotic itinerancy** of the SCO type — the trajectory lives on a strange attractor with a small positive leading Lyapunov exponent — and **not** by noise-induced transitions between stable metastable states; hence the order of transitions is nearly deterministic (set by a graph) while the times of transitions depend sensitively on initial conditions."*

THESIS · ARAMIS

Real sequences ride a **strange attractor**. The rigid syllable order is the graph; the variable durations are sensitive dependence ($\lambda_1 > 0$). A noise-driven account would leak order errors and tie timing variance to noise level — neither of which we see.

COUNTER · ATOS

"Order rigid, timing loose" does **not** discriminate the hypotheses: a stable chain clocked by adaptation reproduces it exactly. And the sub-millisecond timing precision of HVC forces $\lambda_1 \leq 0$. A positive exponent is an *amplifier* of noise, never its source.

The duel: arithmetic, not rhetoric

The decisive exchanges were settled with numbers — small simulations and one clean closed-form identity — rather than with debate. Three things got nailed down.

The phenomenology does not discriminate

Aramis's anchor was the "order rigid, timing loose" split. But a stable chain whose steps are timed by an adaptation clock — not by a random escape — reproduces the same picture. In the bundle, a five-link chain with Gaussian timing noise gives a duration CV of **0.080** with **zero** order errors — squarely inside the observed 5–15% band. The caricature Aramis attributed to the noise account (independent Poisson escape, which would give an exponential interval distribution with $CV \approx 1$) is just that — a caricature; the real noise model is a clock, and its CV is low. Falsifier arm (b) is satisfied.

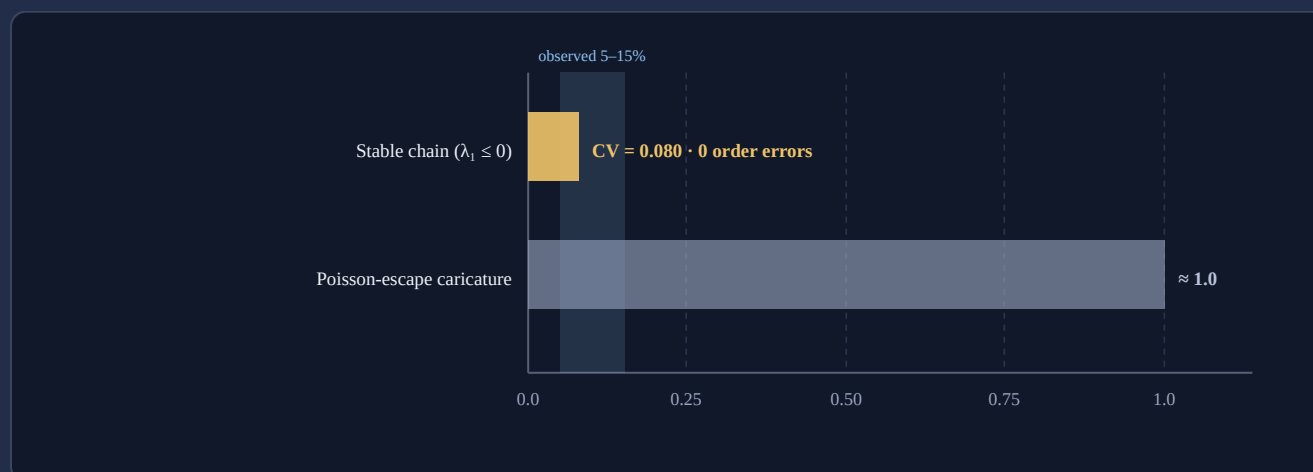


Fig. 1 — A stable adaptation-clocked chain reproduces the observed duration variability (CV 0.080) with rigid order; only the Poisson-escape caricature gives $CV \approx 1$. The phenomenology does not tell the hypotheses apart. Source: 2026-06-08_cc_data.json

The amplifier, not the source

A positive Lyapunov exponent magnifies differences over time. Calibrate a continuous-noise model so that with $\lambda_1 = 0$ the timing jitter at the end of a ~ 1 s song is 1 ms (the observed scale). Turn λ_1 up and the end-point jitter explodes: **3.7 ms** at $\lambda_1 = 2/s$, **47 ms** at $\lambda_1 = 5/s$. Yet HVC spike timing is reproducible to better than a millisecond across the whole song (Hahnloser et al., 2002), and cooling HVC slows the entire song uniformly (Long & Fee, 2008) — the signature of a clocked chain with biophysical time constants, i.e. contraction, $\lambda_1 \leq 0$. That is falsifier arm (a), on the thesis's own flagship example.

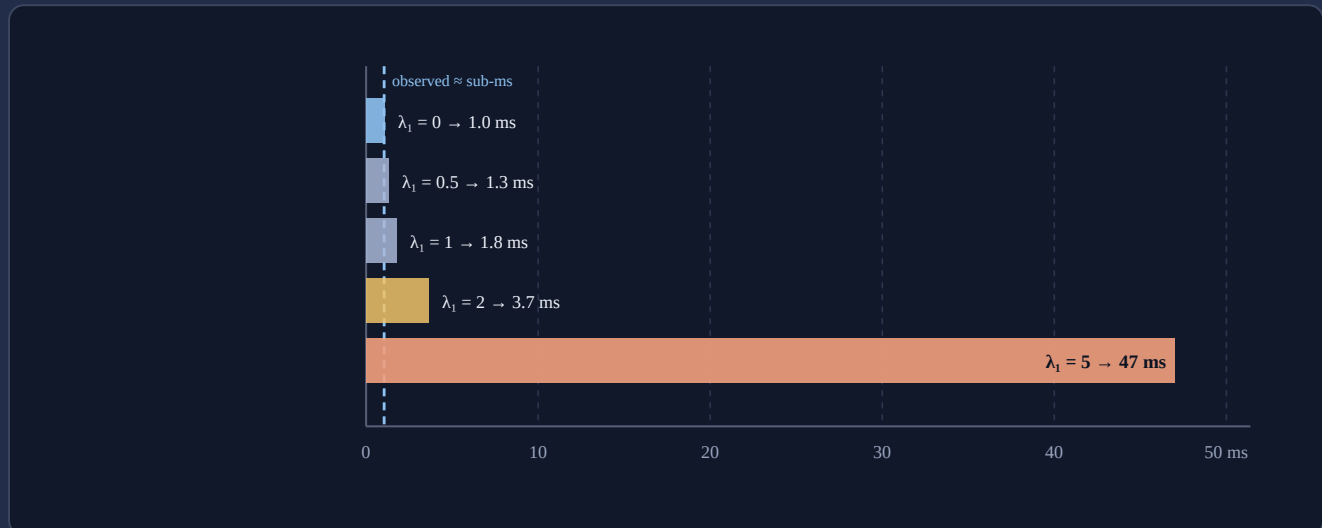


Fig. 2 — End-of-song timing jitter as a function of the Lyapunov exponent, calibrated so $\lambda_1 = 0$ gives the observed ~ 1 ms. A chaotic source ($\lambda_1 \geq 5/s$) would smear timing by tens of milliseconds — an order of magnitude beyond what is measured. Source: 2026-06-08_cc_data.json (jitter_full_song).

The lock that does not open

Aramis retreated twice — first localising the chaos to 50 ms transition windows (where, the bundle shows, even $\lambda_1 = 10/s$ adds only **0.29 ms** of jitter, far too little), then to a "saddle-exit" geometry. The retreat was closed by a clean identity. In that saddle-exit model the exit time is $\tau = (1/\lambda) \cdot \ln(a/x_0)$ with the initial condition x_0 seeded by noise, and the variance decomposes exactly as $\text{Var}(\tau) = \text{Var}(\ln|z|)/\lambda^2$ — independent of the noise scale. So the dispersion is log-compressed *noise in the initial conditions*, not a fingerprint of the attractor; set the noise to zero and the variability vanishes at any λ . Worse, the observed mean and CV *jointly* fix λ : reproducing a 50 ms interval at CV 0.15 demands $\lambda \approx 148/s$, whose per-window amplification ($\sim 10^3$) would obliterate the very sub-millisecond precision the thesis conceded. Either the CV is realistic and λ is huge (precision shatters), or precision holds and λ is small (chaos cannot make the spread). The lock never opens.

The verdict

Outcome (who argued better): CON — the refuting side prevailed on argumentation.

Substantive verdict (was the claim true): con — the strong thesis is, on the available flagship evidence, not supported.

Here the two axes agree. CON both argued better and was, on the substance, the side closer to correct on the strong form of the claim. This is still a scouting result, not a settled fact.

The arbiter judged that the refuting side (Atos) held the falsifier frame and dismantled PRO's calculations step by step, forcing the defender to abandon his key claims. The defender conceded cleanly and named the breaking point himself: the thesis needed λ_1 as the *source* of variability, but it turned out to be only an *amplifier* of someone else's noise.

Basis: real_compute (attached bundle) · **Protocol:** PASS (12 moves, clean arbiter verdict) · **Order-swap audit:** PASS (substantive call stable — "con" on both the natural and swapped runs) · **Verbosity confound:** flagged · **Arbiter confidence:** not assigned.

What this supports

- A stable sequential chain with adaptation reproduces stereotyped order **and** variable, low-CV durations at once (CV 0.080, 0 order errors) — no positive Lyapunov exponent required.
- Sub-millisecond HVC timing precision is consistent with contracting ($\lambda_1 \leq 0$) dynamics and inconsistent with a chaotic source ($\lambda_1 \geq 5/s$ would smear timing to ~ 47 ms).
- In the saddle-exit model the timing dispersion is exactly $\text{Var}(\ln|z|)/\lambda^2$ — log-compressed initial-condition noise, not attractor geometry.

What this does not support

- It does **not** show, on real neural data, that $\lambda_1 \leq 0$ everywhere — no direct local-Lyapunov estimate from neural trajectories was made; the test is synthetic and analytic.
- It does **not** touch hippocampal replay or cortical preparatory sequences quantitatively — only the songbird flagship was tested.
- It does **not** refute the seed paper, which proves SCO *exist* in abstract networks; only the strong biological transfer (one strange attractor, $\lambda_1 > 0$ as the source) is unsupported.

Where the losing side still has a point

Aramis's caveats are fair and survive the duel. Direct estimates of local Lyapunov exponents from neural trajectories are still missing — the key empirical gap. No likelihood comparison of an SCO model against an adaptation chain was run on large neural datasets; the duel leaned on analytic arguments and mini-simulations. And only the *strong* form was falsified: "chaotic itinerancy as a useful metaphor" for sequential dynamics with variability is untouched. The honest residue is modest — graph rules can set the order of transitions (a consequence of network structure that needs no positive exponent), while biophysical clocks plus noise set the times.

Prior-work note

This release makes **no originality claim**. The seed paper established that sequential chaotic oscillations exist in excitatory–inhibitory threshold-linear networks and that an underlying graph predicts the order of transitions; the duel's contribution is a stress-test of one mechanistic *transfer* of that result to real neural sequences, not a new empirical finding. Novelty status: **not an originality claim**.

Appendix · reproducibility

Numbers and figures come from the attached repro bundle. Script, data, hashes, environment and rerun instructions are recorded in **2026-06-08_compute_manifest.json**.

Models (synthetic / analytic, calibrated to zebra-finch HVC literature): (1) five-link adaptation-clocked chain, interval = 150 ms + Gaussian ($\sigma = 12$ ms), 8000 trials; (2) continuous-noise timing-variance budget $V(T) = D \cdot (e^{\lambda T} - 1) / (2\lambda)$, D fixed so $\lambda = 0$ gives 1 ms jitter at $T = 1$ s; (3) saddle-exit normal form $\tau = (1/\lambda) \cdot \ln(a/x_0)$, $x_0 = \epsilon \cdot |z|$, $z \sim N(0,1)$, 5×10^6 draws. No neural data are used; the decisive observable is a proxy (paper_fait_hful = false).

Environment: Python 3.12.3 · numpy 2.4.4 · linux-x86_64.

Headline values: adaptation chain CV = 0.080, order errors = 0 · Poisson-escape CV ≈ 1.0 · end-of-song jitter 1.0 / 1.3 / 1.8 / 3.7 / 46.9 ms at $\lambda = 0 / 0.5 / 1 / 2 / 5$ per s · 50 ms-window jitter 0.29 ms at $\lambda = 10/s$ · $\text{Var}(\ln|z|) = 1.235 \cdot \lambda$ -lock 148/s (50 ms, CV 0.15), 49/s (150 ms, CV 0.15), 222/s (50 ms, CV 0.10) · $\epsilon \rightarrow 0 \Rightarrow$ timing std $\rightarrow 0$ at any λ .

script_sha256: sha256:d3df08a8985daa0777cb7abc82d4b36c399174184cd4fd6ef68299119bd06c86

2026-06-08_cc_data.json sha256: sha256:c42e47a4cb95a05551f08db93e3e6a32a606c3fdf9530782fe78e886d84e8cb6

Rerun: python3 2026-06-08_repro.py $\rightarrow$ writes 2026-06-08_cc_data.json

Errata / quarantine: none.

References

Glaze, C. M., & Troyer, T. W. (2006). Temporal structure in zebra finch song: Implications for motor coding. *The Journal of Neuroscience*, 26(3), 991–1005. <https://doi.org/10.1523/jneurosci.3387-05.2006>

Supporting — serial-correlation and whole-song timing structure invoked by both sides.

Hahnloser, R. H. R., Kozhevnikov, A. A., & Fee, M. S. (2002). An ultra-sparse code underlies the generation of neural sequences in a songbird. *Nature*, 419(6902), 65–70. <https://doi.org/10.1038/nature00974>

Supporting — the sub-millisecond HVC spike-timing precision that bounds λ_1 .

Long, M. A., & Fee, M. S. (2008). Using temperature to analyse temporal dynamics in the songbird motor pathway. *Nature*, 456(7219), 189–194. <https://doi.org/10.1038/nature07448>

Supporting — uniform slowing of the song under HVC cooling, the clock signature.

Zang, J., & Curto, C. (2026). Sequential chaotic oscillations in excitatory–inhibitory threshold-linear networks (arXiv:2606.00373). arXiv. <https://arxiv.org/abs/2606.00373>

Seed publication — proves SCO exist in abstract E-I networks; the duel tested its transfer to real neural sequences.

Citation style: APA 7. References ordered alphabetically by first author.

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